

Recurrent Neural Networks

This lecture

- ❖ Review: Neural Network
- ❖ Recurrent NN
 - ❖ GRU
 - ❖ LSTM
 - ❖ Bidirectional RNN

Examples of Sequence Data

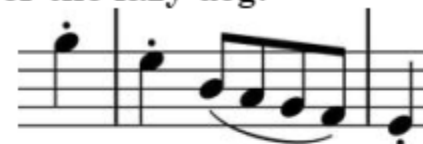
Speech recognition



"The quick brown fox jumped over the lazy dog."

Music generation

∅



Sentiment classification

"There is nothing to like in this movie."



DNA sequence analysis

AGCCCCTGTGAGGAACTAG



AG**CCCCTGTGAGGAACTAG**

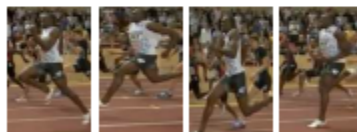
Machine translation

Voulez-vous chanter avec moi?



Do you want to sing with me?

Video activity recognition



Running

Name entity recognition

Yesterday, Harry Potter met Hermione Granger.



Yesterday, **Harry Potter** met **Hermione Granger**.

A sample Problem

Harry Potter and Hermione Granger invented a new spell

$\langle x^1 \rangle$ $\langle x^2 \rangle$ $\langle x^3 \rangle$ $\langle x^4 \rangle$ $\langle x^5 \rangle$ $\langle x^t \rangle$ $\langle x^9 \rangle$

$\langle y^1 \rangle$ $\langle y^2 \rangle$ $\langle y^3 \rangle$ $\langle y^4 \rangle$ $\langle y^5 \rangle$ $\langle y^t \rangle$ $\langle y^9 \rangle$

1 1 0 1 1 0 0

Word Representation

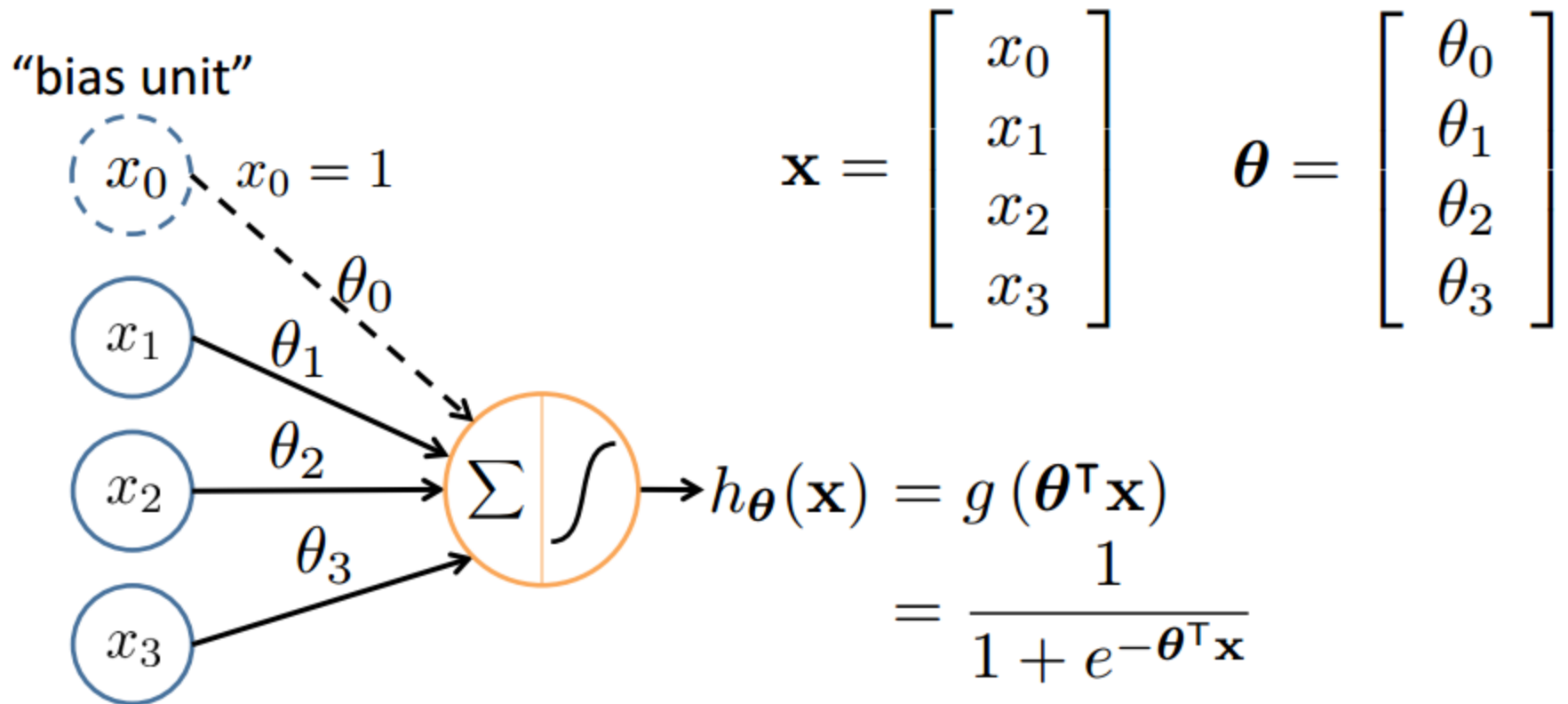
Harry Potter and Hermione Granger invented a new spell

$\langle x^1 \rangle$ $\langle x^2 \rangle$ $\langle x^3 \rangle$ $\langle x^4 \rangle$ $\langle x^5 \rangle$ $\langle x^t \rangle$ $\langle x^9 \rangle$

One hot encoding

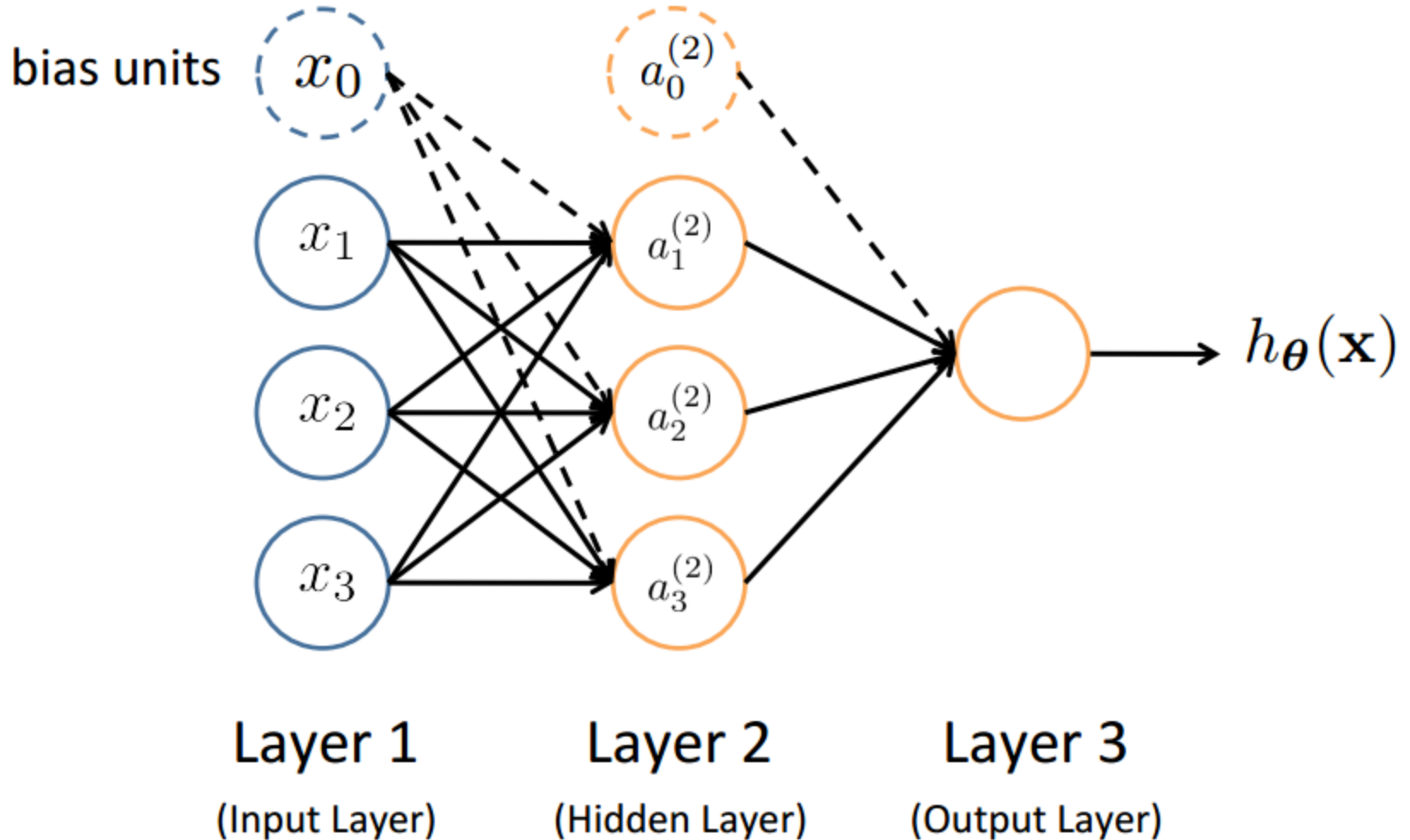
size of the vector is same size of the vocabulary

Neural Network



Sigmoid (logistic) activation function: $g(z) = \frac{1}{1 + e^{-z}}$

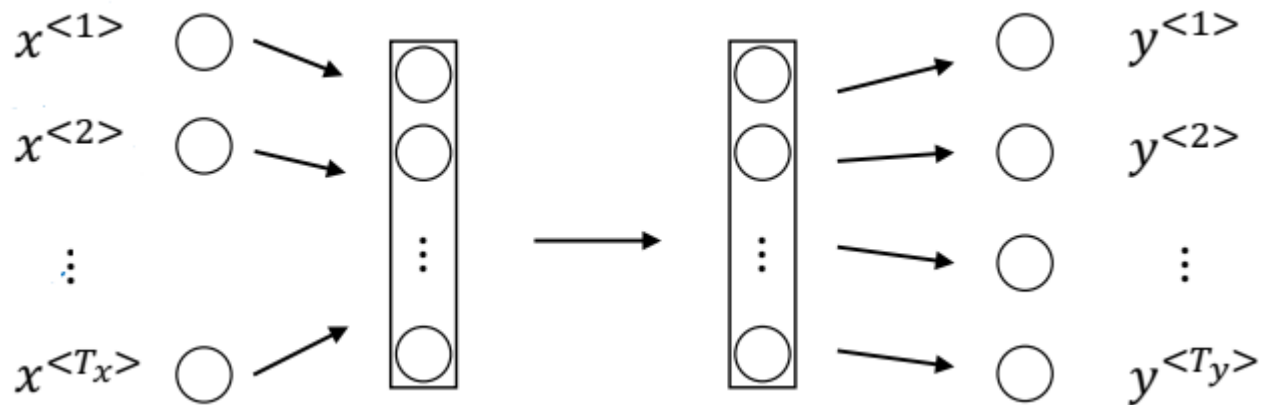
Neural Network (feed forward)



Why standard NN is not Fit for Sequence Data?

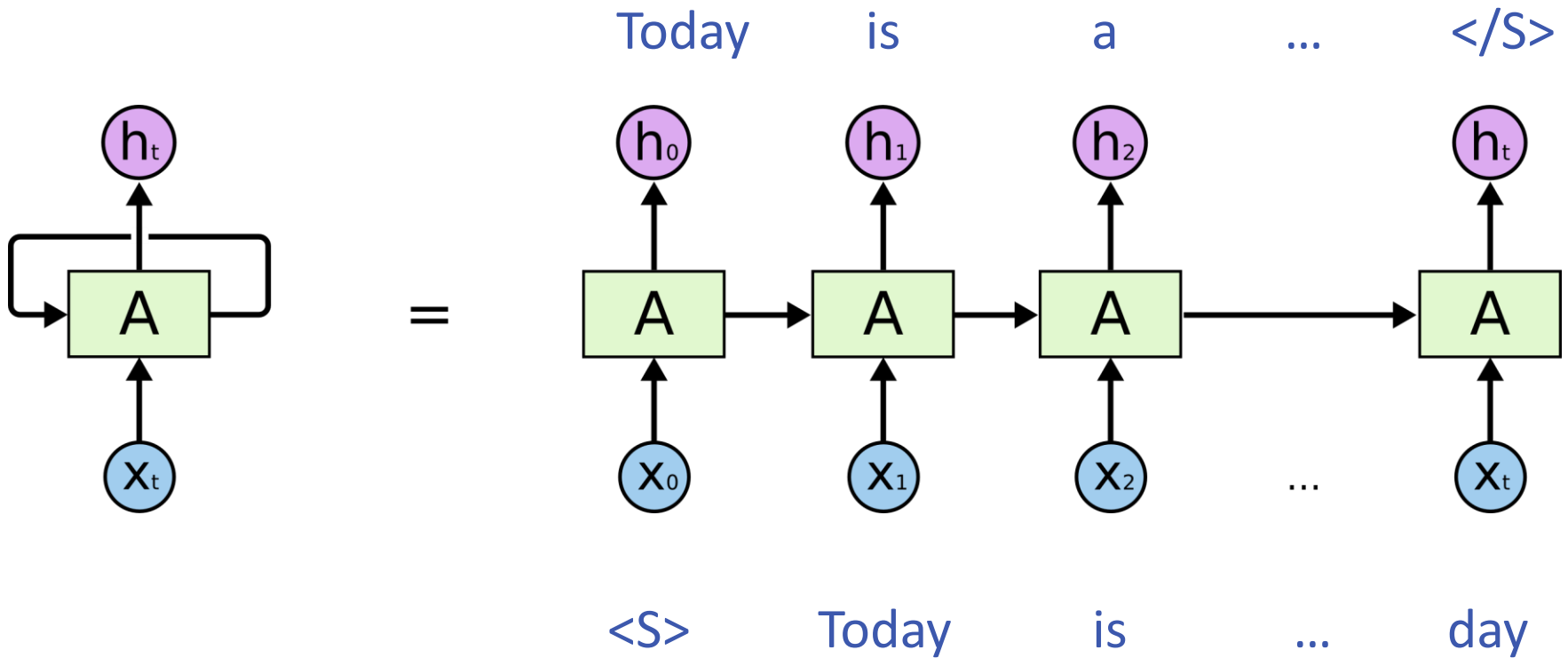
❖ Problems

- ❖ Input and output can be of different lengths for different examples
- ❖ No feature sharing learned for different positions of text



How to deal with input with variant size?

❖ Use same parameters

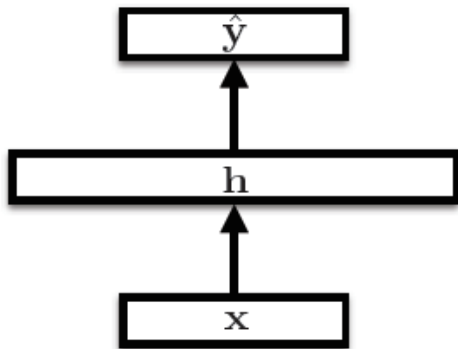


Recurrent Neural Networks

Feed-forward NN

$$\mathbf{h} = g(\mathbf{V}\mathbf{x} + \mathbf{c})$$

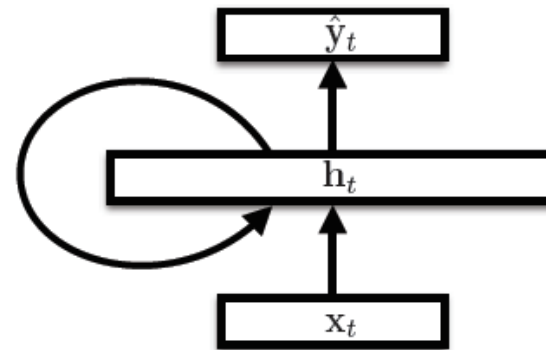
$$\hat{\mathbf{y}} = \mathbf{W}\mathbf{h} + \mathbf{b}$$



Recurrent NN

$$\mathbf{h}_t = g(\mathbf{V}\mathbf{x}_t + \mathbf{U}\mathbf{h}_{t-1} + \mathbf{c})$$

$$\hat{\mathbf{y}}_t = \mathbf{W}\mathbf{h}_t + \mathbf{b}$$



Example

•**Vocabulary:** ["I", "love", "AI", "chat"]

•**One-Hot Encodings:**

•"I": [1, 0, 0, 0]

•"love": [0, 1, 0, 0]

•"AI": [0, 0, 1, 0]

•"chat": [0, 0, 0, 1]

$$V = \begin{bmatrix} 0.2 & 0.3 & 0.5 & 0.7 \\ 0.4 & 0.1 & 0.6 & 0.8 \end{bmatrix}, U = \begin{bmatrix} 0.1 & 0.3 \\ 0.4 & 0.2 \end{bmatrix}, c = \begin{bmatrix} 0.1 \\ 0.1 \end{bmatrix}$$

$$W = \begin{bmatrix} 0.6 & 0.9 \\ 0.7 & 0.2 \end{bmatrix}, b = \begin{bmatrix} 0.1 \\ 0.2 \end{bmatrix}$$

$$h_0 = \begin{bmatrix} 0 \\ 0 \end{bmatrix}, x_1 = \begin{bmatrix} 1 \\ 0 \\ 0 \\ 0 \end{bmatrix} \text{ and } h_t = \tanh(V * x_t + U * h_{t-1} + c)$$

$$h_1 = \begin{bmatrix} 0.29 \\ 0.46 \end{bmatrix}, h_2 = \begin{bmatrix} 0.51 \\ 0.39 \end{bmatrix}, h_3 = \begin{bmatrix} 0.65 \\ 0.75 \end{bmatrix}, h_4 = \begin{bmatrix} 0.79 \\ 0.86 \end{bmatrix},$$

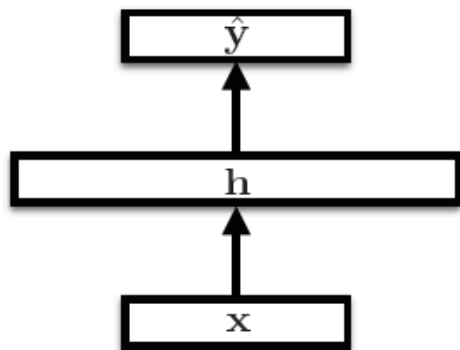
$$\hat{y}_1 = \begin{bmatrix} 0.688 \\ 0.495 \end{bmatrix}, \hat{y}_2 = \begin{bmatrix} 0.757 \\ 0.635 \end{bmatrix}, \hat{y}_3 = \begin{bmatrix} 1.165 \\ 0.805 \end{bmatrix}, \hat{y}_4 = \begin{bmatrix} 1.348 \\ 0.925 \end{bmatrix}$$

Recurrent Neural Networks

Feed-forward NN

$$\mathbf{h} = g(\mathbf{V}\mathbf{x} + \mathbf{c})$$

$$\hat{\mathbf{y}} = \mathbf{W}\mathbf{h} + \mathbf{b}$$

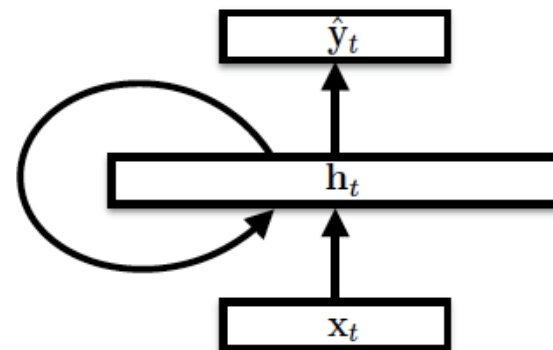


Recurrent NN

~~$$\mathbf{h}_t = g(\mathbf{V}\mathbf{x}_t + \mathbf{U}\mathbf{h}_{t-1} + \mathbf{c})$$~~

$$\mathbf{h}_t = g(\mathbf{V}[\mathbf{x}_t; \mathbf{h}_{t-1}] + \mathbf{c})$$

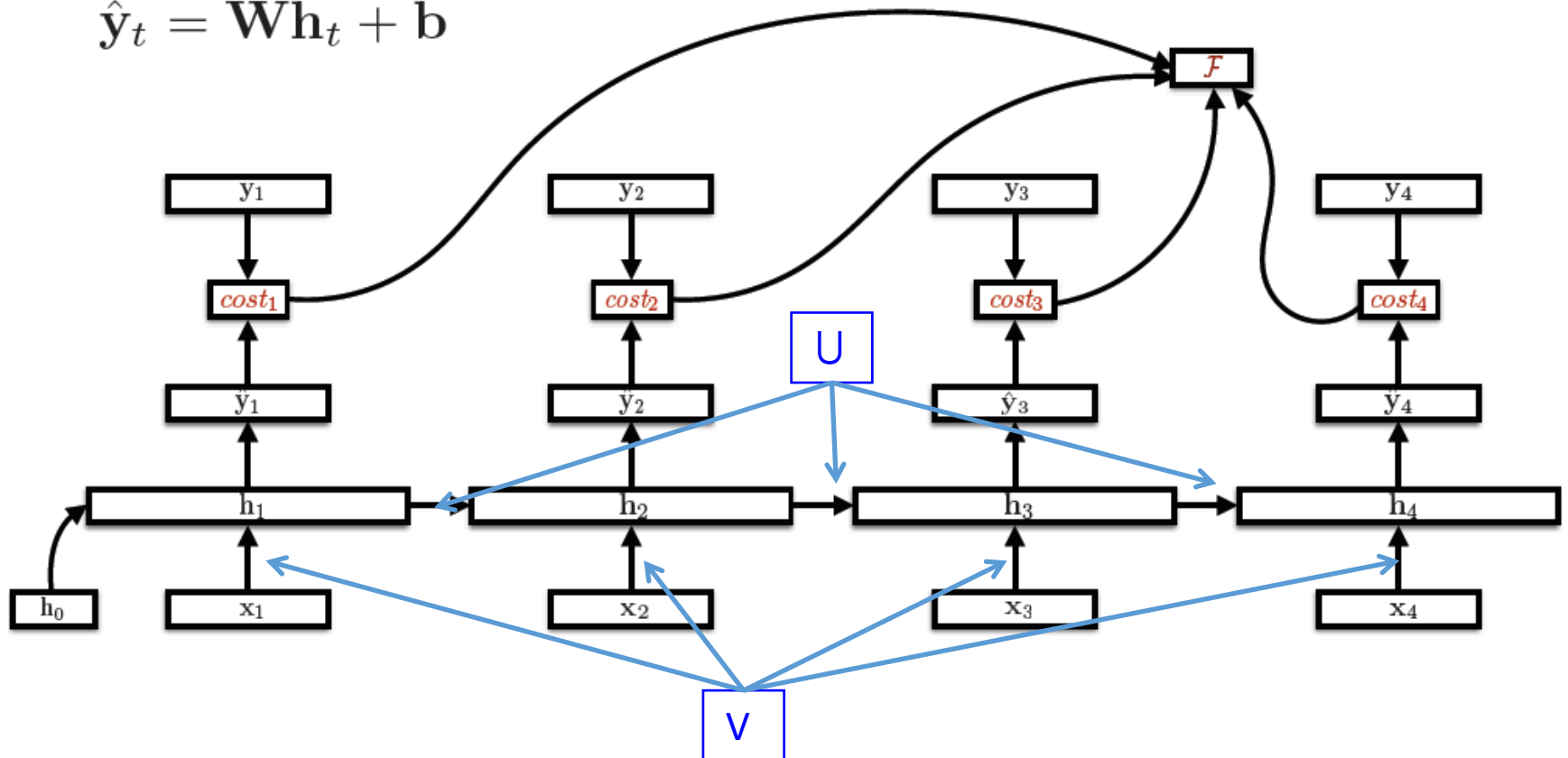
$$\hat{\mathbf{y}}_t = \mathbf{W}\mathbf{h}_t + \mathbf{b}$$



Unroll RNNs

$$\mathbf{h}_t = g(\mathbf{V}\mathbf{x}_t + \mathbf{U}\mathbf{h}_{t-1} + \mathbf{c})$$

$$\hat{\mathbf{y}}_t = \mathbf{W}\mathbf{h}_t + \mathbf{b}$$

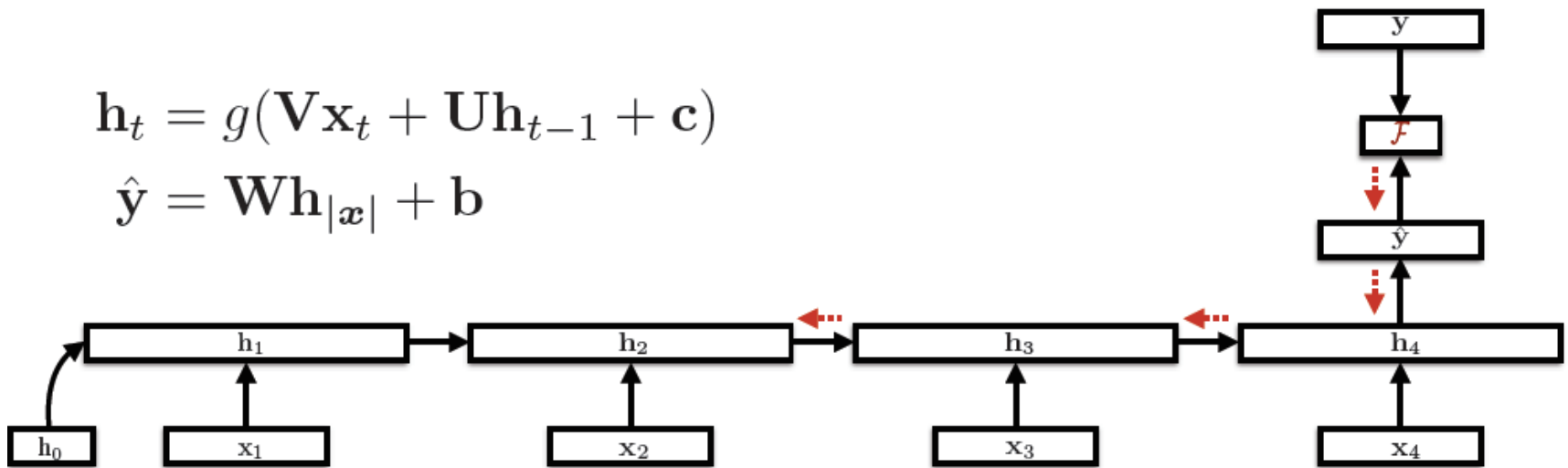


RNN training

❖ Back-propagation over time

$$\mathbf{h}_t = g(\mathbf{V}\mathbf{x}_t + \mathbf{U}\mathbf{h}_{t-1} + \mathbf{c})$$

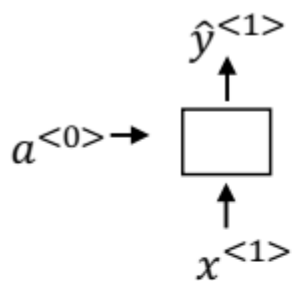
$$\hat{\mathbf{y}} = \mathbf{W}\mathbf{h}_{|\mathbf{x}|} + \mathbf{b}$$



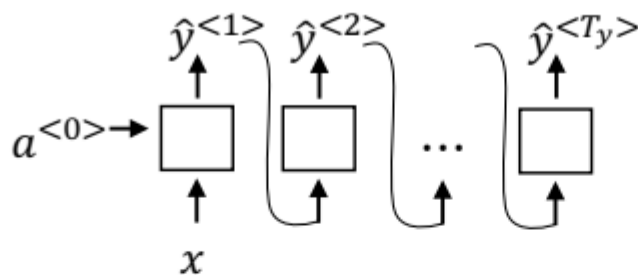
What happens to gradients as you go back in time?

$$\frac{\partial \mathbf{h}_3}{\partial \mathbf{h}_2} \frac{\partial \mathbf{h}_4}{\partial \mathbf{h}_3} \frac{\partial \hat{\mathbf{y}}}{\partial \mathbf{h}_4} \frac{\partial \mathcal{F}}{\partial \hat{\mathbf{y}}} \frac{\partial \mathcal{F}}{\partial \mathcal{F}}$$

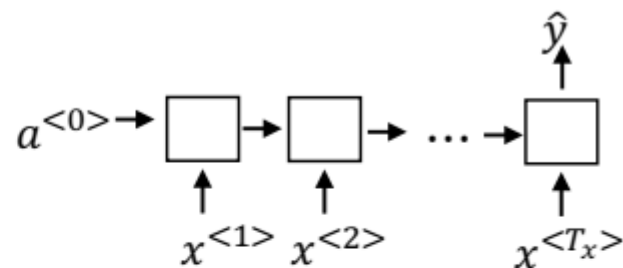
Different Types of RNNs



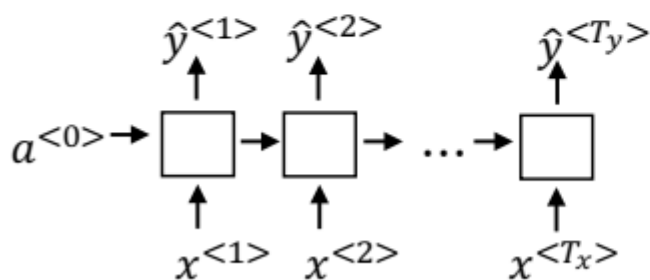
One to one



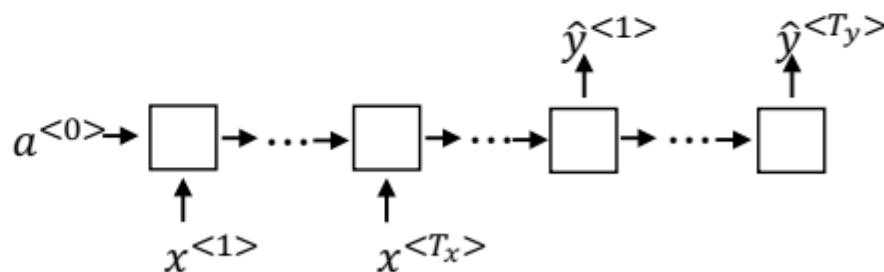
One to many



Many to one



Many to many



Many to many

Language Model and Sequence Generation

What is Language modelling?

Speech Recognition

The apple and pair salad

The apple and pear salad

$$P(\text{The apple and pair salad}) = 7.3 \times 10^{-13}$$

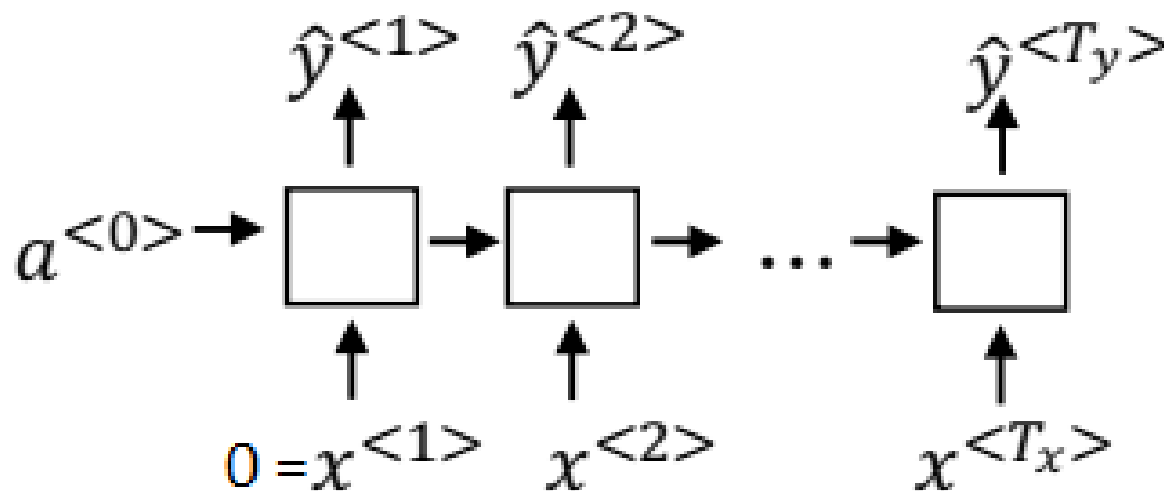
$$P(\text{The apple and pear salad}) = 3.9 \times 10^{-7}$$

$$P(\text{sentence}) = ?$$

$$P(y^1, y^2, \dots, y^t)$$

Language Modelling with RNN

- ❖ Training set: Large corpus of English text
- ❖ Tokenize sentences
- ❖ Add unknown and EOS in the vocabulary



Vanishing Gradients

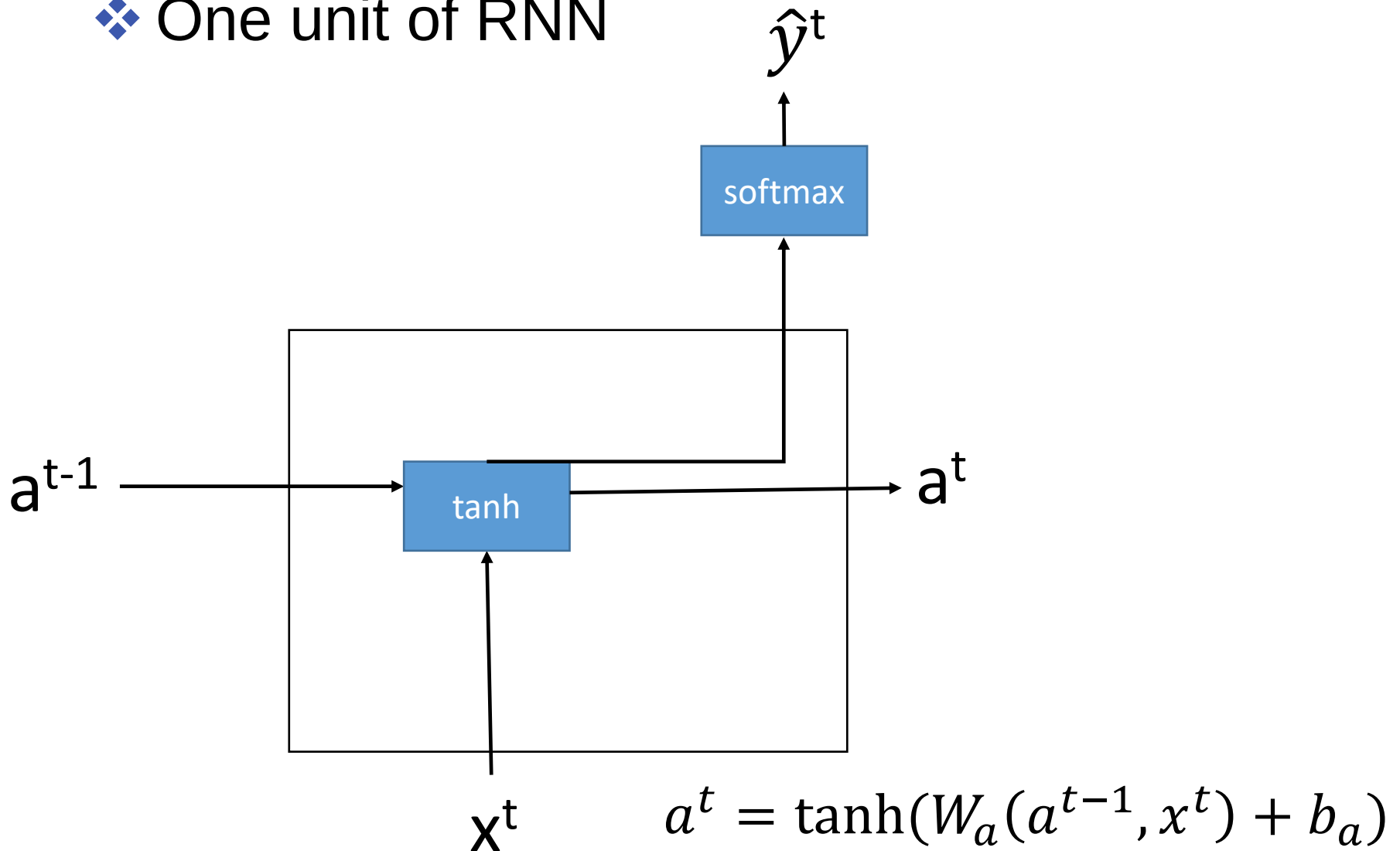
- ❖ For the traditional activation functions, each gradient term has the value in range $(-1, 1)$.
- ❖ Multiplying n of these small numbers to compute gradients
- ❖ The longer the sequence is, the more severe the problems are.

RNNs characteristics

- ❖ Model hidden states (input) dependencies
- ❖ Errors “back propagation over time”
- ❖ **Vanishing gradient** problem:
cannot model long-distant dependencies of the hidden states.

Gated Recurrent Unit (GRU)

❖ One unit of RNN



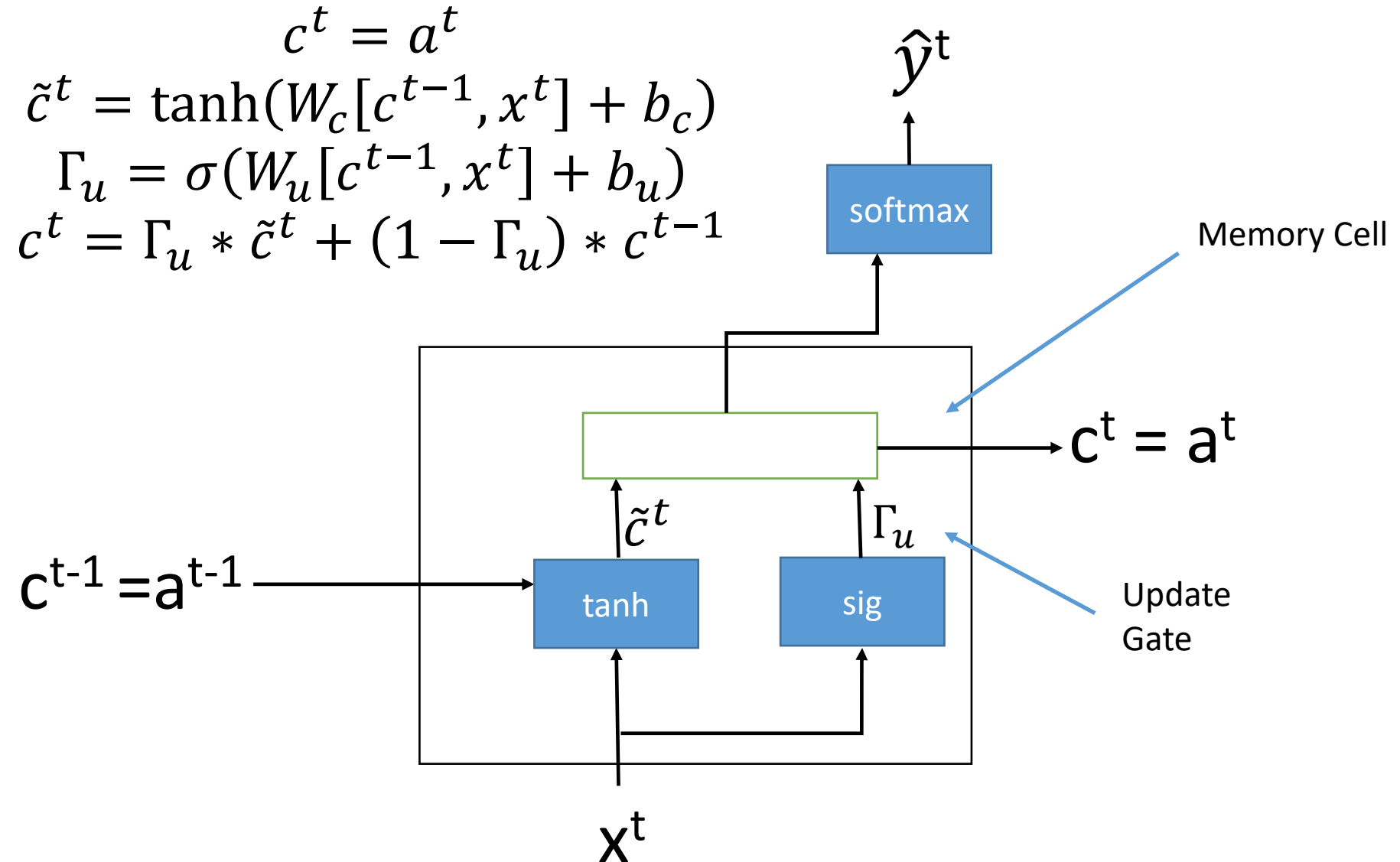
GRU (Simplified)

$$c^t = a^t$$

$$\tilde{c}^t = \tanh(W_c[c^{t-1}, x^t] + b_c)$$

$$\Gamma_u = \sigma(W_u[c^{t-1}, x^t] + b_u)$$

$$c^t = \Gamma_u * \tilde{c}^t + (1 - \Gamma_u) * c^{t-1}$$



GRU Full

$$c^t = a^t$$

$$\tilde{c}^t = \tanh(W_c[\Gamma_r * c^{t-1}, x^t] + b_c)$$

$$\Gamma_u = \sigma(W_u[c^{t-1}, x^t] + b_u)$$

$$\Gamma_r = \sigma(W_r[c^{t-1}, x^t] + b_r)$$

$$c^t = \Gamma_u * \tilde{c}^t + (1 - \Gamma_u) * c^{t-1}$$

Long-Short Term Memory Networks (LSTMs)

$$\tilde{c}^{<t>} = \tanh(W_c[a^{<t-1>}, x^{<t>}] + b_c)$$

$$\Gamma_u = \sigma(W_u[a^{<t-1>}, x^{<t>}] + b_u)$$

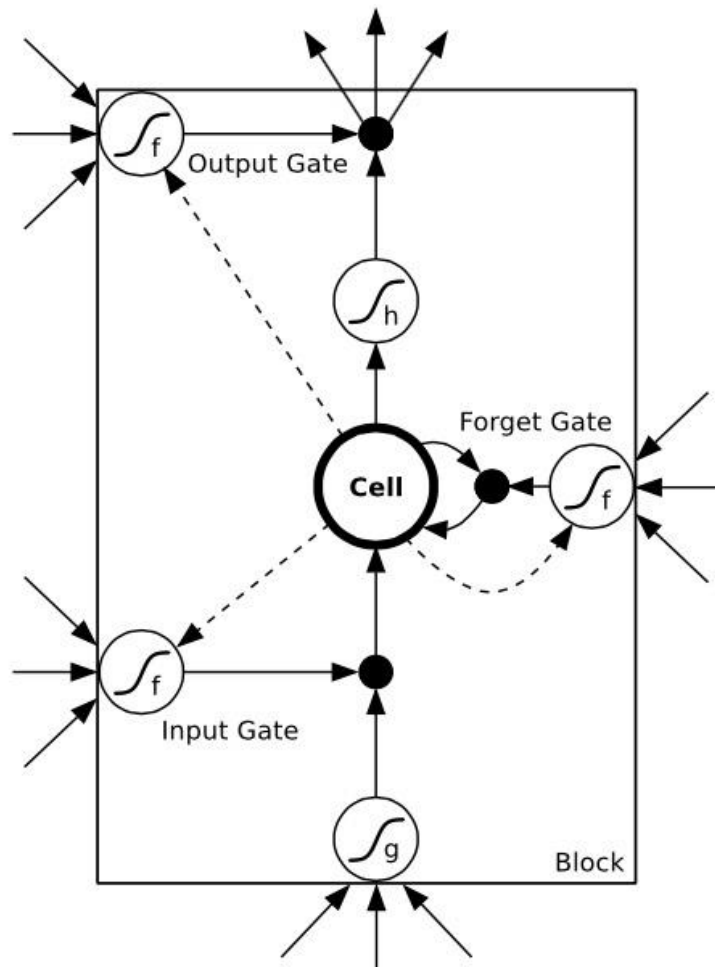
$$\Gamma_f = \sigma(W_f[a^{<t-1>}, x^{<t>}] + b_f)$$

$$\Gamma_o = \sigma(W_o[a^{<t-1>}, x^{<t>}] + b_o)$$

$$c^{<t>} = \Gamma_u * \tilde{c}^{<t>} + \Gamma_f * c^{<t-1>}$$

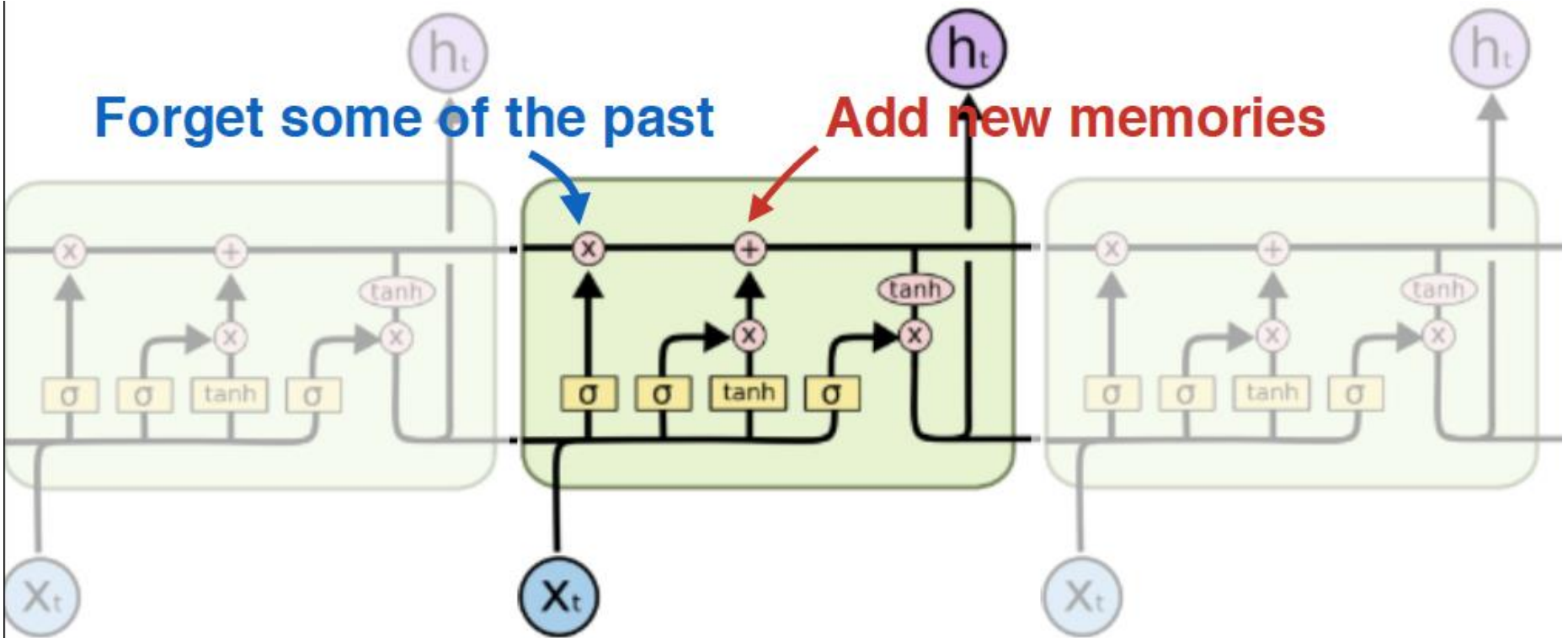
$$a^{<t>} = \Gamma_o * c^{<t>}$$

Long-Short Term Memory Networks (LSTMs)



Use gates to control the information to be added from the **input**, **forgot** from the previous memories, and **outputted**. σ and f are *sigmoid* and *tanh* function respectively, to map the value to $[-1, 1]$

Another Visualization



Capable of modeling long-distant dependencies between states.

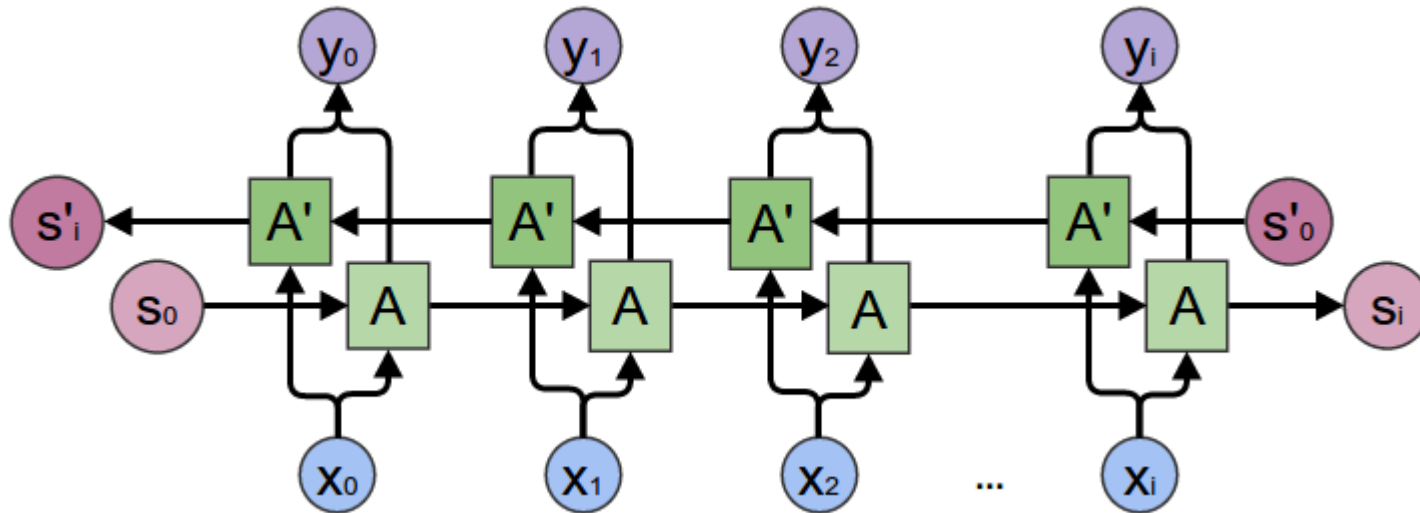
Figure credit: Christopher Olah

Bidirectional RNN

Getting information from the future

He said, “Teddy bears are on sale!”

He said, “Teddy Roosevelt was a great President!”



Deep RNN

